Behavior-driven Hypergraph Modeling for Accurate Cyberbullying Detection in Social Media

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***Abstract*—With the widespread adoption of social media and the growing complexity of online interactions, cyberbullying has increasingly exhibited characteristics such as group involvement, concealment, and coordinated behaviors. Most existing approaches model interactions by using linear sequences or binary graphs, which primarily capture simple local relationships but struggle to mimic multi-user coordinated attacks effectively. Moreover, these methods fail to model the relationship among group behaviors that naturally emerge during the evolution of cyberbullying. To address these challenges, we propose a Behavior-driven Hypergraph Model for Cyberbullying Detection (BeHM), which parses cyberbullying behaviors within media sessions at two complementary levels: coordinated interactions at the micro-level, and their relations at the macro-level. Then, BeHM adopts a three-stage framework to capture the features at the micro and macro levels. Experiments on two public session-based social media datasets demonstrate that BeHM consistently outperforms baselines, achieving an F1-score with improvements of up to 1.3% on Instagram and 2.8% on Vine, and validating its effectiveness in modeling complex group attacks and relationships among group behaviors.**

***Index Terms*—cyberbullying detection, social media analysis, hypergraphs**

I. Introduction

Defined as repeated aggressive behaviors carried out through digital devices against vulnerable targets, cyberbullying has become a global issue with the rapid growth of social media [1]. Approximately half of young people have experienced some forms of cyberbullying, and 10-20% faced it regularly [2]. Victims often suffer from depression, anxiety, and decreased self-esteem; in severe cases, they may even exhibit self-harm or suicidal tendencies [3]. Therefore, effective cyberbullying detection is critically important.

Recent studies have leveraged Graph Neural Networks (GNNs) [4] to model media sessions through reply-to or user interaction graphs [5], [6]. However, these pairwise graph-based methods primarily capture individual user behaviors (e.g., user A attacks user B), but struggle to model the coordinated group behavior [7], [8], which manifests as collective encirclement cyberbullying behaviors formed through interactions among multiple users, as shown in Fig. 1. Moreover, existing methods treat a group behavior as an isolated behavior [9], [10], which neglects the relationship among different group behaviors. E.g., a few groups attack the same victim, which causes these groups to support each other stronger and stronger, and creates a sustained harassment environment for the victims. These limitations in modeling coordinated group behaviors and their relationship severely restrict the identification of covert and persistent cyberbullying.

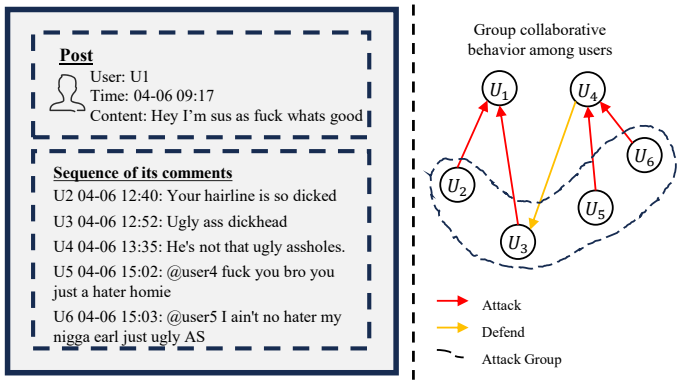


Fig. 1 Illustration of collective encirclement cyberbullying behaviors. The left shows an example of a cyberbullied media session with a post and sequential comments, while the right presents the corresponding interaction pattern, which illustrates coordinated attacks from multiple users (U_2, U_3, U_5, and U_6) against the post author U_1 and user U_4.

As an extension of traditional graphs, hypergraphs leverage their unique hyperedges to connect multiple nodes simultaneously and thus offer a more natural modeling framework for describing coordinated behaviors [11]. These structures have been widely applied in areas such as fake news detection and hate speech detection. However, most existing studies [12], [13] primarily model macroscopic relationships among sessions rather than providing an in-depth analysis of the complex interactions within each session.

To address these challenges, we propose a Behavior-driven Hypergraph Model for Cyberbullying Detection (BeHM), which captures cyberbullying through two complementary levels. At the micro-level, we model and connect related multi-user coordinated interactions with a hyperedge that represents an individual group behavior, and at the macro-level, an Interaction Behavior Graph (IBG) captures different relations among group behaviors, and then, related relations are formed as communities through a community discovery method. Through a top-down fusion strategy, BeHM integrates the features at these two complementary levels and develops an effective identification of concealed and persistent cyberbullying behaviors. Overall, the main contributions of this study

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are summarized as follows:

- We highlight the importance of modeling coordinated group behaviors for cyberbullying detection and present a hypergraph to explicitly capture collective interactions within sessions through multiple types of hyperedges.
- We propose a behavior-driven hypergraph framework that treats hyperedges as coordinated group behaviors and further converts them into an Interaction Behavior Graph, for better capturing the relationships among group behaviors at a macro level.
- We design a multi-dimensional fusion mechanism that aligns the features at these two complementary levels, improving robustness and interpretability for session-level cyberbullying detection.

The proposed BeHM model is evaluated on two public datasets, Instagram [14] and Vine [15]. The comparative results demonstrate that the BeHM model outperforms the existing state-of-the-art models and comprehensively verifies its effectiveness in complex cyberbullying detection scenarios.

II. Related Work

A. Cyberbullying Detection

Over the past decade, the field of cyberbullying detection has evolved from sequence modeling to graph modeling. Early work models sessions as sequential data by using hierarchical attention networks [16]. Cheng et al. [9] extend this with bidirectional GRUs and hierarchical attentions to capture word-level and comment-level contexts. Recent studies shift the focus to user interactions by using Graph Neural Networks (GNNs) [4]. Chen and Li [5] employ Graph Convolutional Networks (GCNs) to capture inter-session and inter-post connections, while Ge et al. [6] model the evolving interactions by using a time graph. Building on this, Zhou et al. [17] introduce TUHIN to model multidimensional social relationships via heterogeneous graphs and temporal attention. However, traditional graphs are limited by binary connectivity and struggle to capture group-based cyberbullying with multiple participants (attackers, victims, bystanders). When GNNs model collective attacks involving three or more users, their message-passing mechanism suffers from information dilution. It creates low-order representation bottlenecks, which weakens their ability to handle complex group interactions.

B. Hypergraph Learning

Hypergraphs attract widespread attention due to their natural ability to model complex high-order relationships. Hypergraph Neural Networks (HGNNs) [18] explicitly capture high-order relationships by constructing hyperedges among semantically similar nodes based on distance metrics. The attention mechanism for hypergraph introduced on this basis further enhances the model's power for expression. HyperGAT [19] employs a two-level attention mechanism to capture high-order semantic dependencies between words and sentences in text classification tasks. Hypergraph neural networks are widely applied in social media analysis. Jeong et al. [12] propose a framework that constructs three types of hyperedges (user, time, and entity) to capture group collaboration relationships during rumor dissemination. Su and Yang [13] construct hypergraphs for user attributes to represent non-textual high-order relationships in the user participation for the news dissemination. Xu et al. [20] propose a temporal hypergraph neural network for information diffusion modeling, which captures both local interactions and global diffusion patterns by jointly modeling high-order interactions and their temporal dynamics. These studies demonstrate the significant potential of hypergraphs in capturing diverse interactions.

III. Problem Statement

Let $O = \{S_1, S_2, \ldots, S_N\}$ denote a dataset consisting of N social media sessions for analysis, where S_i corresponds to the i-th session. Each session S_i is characterized by a quadruple $S_i = \{P_i, C_i, U_i, Y_i\}$, which includes the initial post P_i, a collection of comments $C_i = \{C_i^1, C_i^2, \ldots, C_i^n\}$, the corresponding users $U_i = \{U_i^P, U_i^{C_1}, U_i^{C_2}, \ldots, U_i^{C_n}\}$ of the post and comments, and its associated ground truth Y_i.

Our objective is to model a session S_i using a behavior-driven hypergraph framework that captures coordinated group behaviors at the micro level and the relationship among these group behaviors at the macro level. By learning multi-level representations that encode high-order interactions and their structural associations, we aim to classify each session as either bullying ($Y_i = 1$) or non-bullying ($Y_i = 0$).

IV. Methodology

The overall framework of BeHM is illustrated in Fig. 2. It consists of three core modules: 1) Hypergraph Construction, which constructs a hypergraph for each session; 2) High-Order Interaction Modeling, which treats hyperedges as coordinated group behaviors and captures relationships among behaviors from a macroscopic perspective; 3) Multi-dimensional Information Fusion, which integrates information across multiple levels to determine whether a session involves cyberbullying. The following provides a detailed elaboration on BeHM.

A. Hypergraph Construction

Initial Feature Extraction. We concatenate the original post P and its comments C in a social media session S into an ordered node sequence $N = \{n_1, \ldots, n_k\}$ as input to BeHM. A pre-trained language model $\mathcal{M}$ (e.g., BERT) is employed to encode each node:

$$X_i = \mathcal{M}(n_i), \quad \forall n_i \in N \tag{1}$$

where $X_i \in \mathbb{R}^d$ denotes the d-dimensional representation of node n_i. These features are then fed into the hypergraph modeling stage.

Hypergraph Structure Modeling. Cyberbullying in social networks is often collaboratively conducted by multiple users, involving behaviors such as collective agreement, siege, or defense [1]. Traditional graphs with binary edges struggle to capture the collective intent and multi-user interactions. We model each session S_i as a hypergraph $H = \{N, E_H\}$, where N is the set of nodes and E_H is the set of hyperedges. To

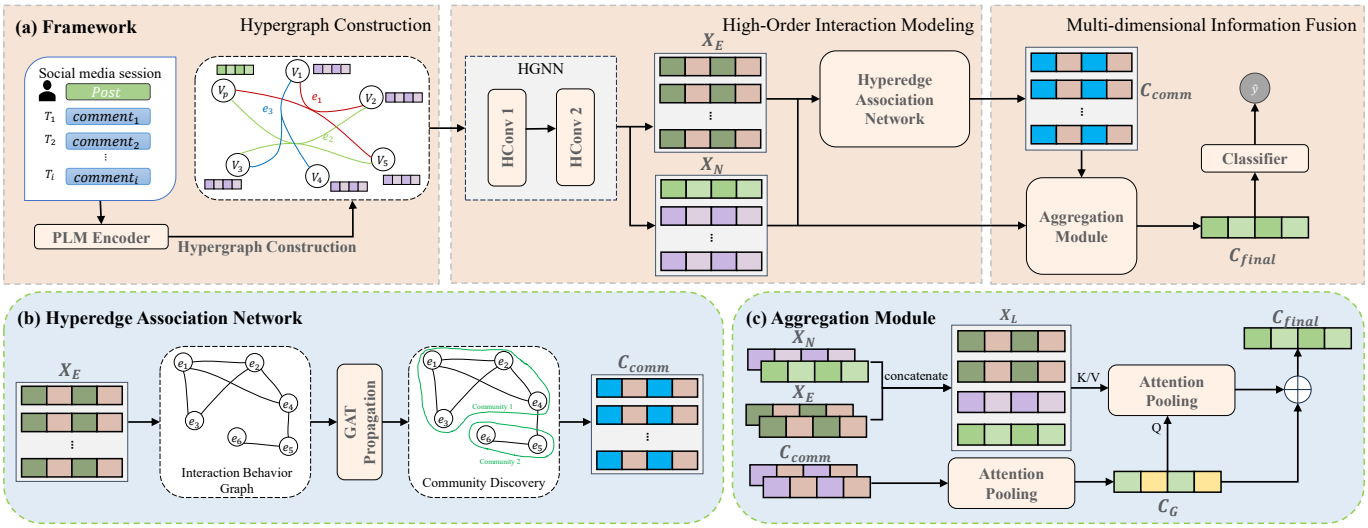


Fig. 2 The framework of BeHM. (a) Overview of the proposed Behavior-driven Hypergraph Model for Cyberbullying Detection. A social media session is modeled as a hypergraph with coordinated group behaviors as hyperedges. Node features X_N, and hyperedge features X_E are learned via hypergraph modeling, while collaborative interaction patterns among group behaviors are captured through interaction graph modeling and are finally fused for session-level classification. (b) Hyperedge Association Network models relationships among group behaviors by constructing an Interaction Behavior Graph and extracting global collaborative patterns via attention and community discovery. (c) Aggregation Module integrates node, hyperedge, and community-level features through multi-dimensional attention to obtain the final session representation.

capture diverse interaction patterns, we define three types of hyperedges, each corresponding to a group behavior.

a) Semantic-Temporal Coordinated Behavior: Existing temporal-based methods often introduce irrelevant nodes, while semantic-based methods fail to capture temporally continuous collective attacks with weak semantic correlations [21]. To address this limitation, we jointly model semantic similarity and temporal contiguity to construct hyperedges that capture implicit coordinated attack behaviors [6]. Specifically, we use the cosine similarity s_{ij} between nodes n_i and n_j to quantify their semantic relevance and define the connection strength between nodes as a modified form of s_{ij}:

$$F_{ij} = s_{ij} \cdot \left(1 + \beta \cdot G(s_{ij}) \cdot \exp\left(-\frac{T_{ij}}{\tau}\right)\right), \tag{2}$$

where d_{ij} denotes the time interval between nodes n_i and n_j, β controls the influence of temporal information, and τ determines temporal sensitivity. To preserve semantic dominance, a binary gate $G(s_{ij})$ activates temporal correction only when $s_{ij} > \eta$. We compare the linking strength F_{ij} between node n_i and node n_j. If F_{ij} is greater than the hyperedge threshold θ, the node pair is added to the hyperedge $E_{STC,i}$.

b) User Coherence Behavior: Cyberbullying sessions often involve users exhibiting consistent attacking or defending behaviors across multiple contents [22]. To capture this behavioral continuity, we construct user coherence hyperedges by aggregating all nodes authored by the same user:

$$E_{User,u} = \{n_i \in N \mid U(n_i) = u\}, \tag{3}$$

where u represents a specific user, and $U(n_i)$ represents the user corresponding to node n_i.

c) Mention Interaction Behavior: @-mentions indicate strong and targeted interactions in social media sessions [23], often reflecting focused group behaviors centered around specific individuals. For each mentioned user u, we define a mention interaction hyperedge that consists of all nodes that mention u and all nodes authored by u.

$$\begin{aligned} E_{Ment,u} = &\{n_j \in N \mid u \in \text{Ment}(n_j)\} \\ &\cup \{n_k \in N \mid U(n_k) = u\}, \end{aligned} \tag{4}$$

where u is a user mentioned in the session, $\text{Ment}(n_j)$ denotes the set of users mentioned in content node n_j.

B. High-Order Interaction Modeling

After constructing the hypergraph, we apply a Hypergraph Neural Network (HGNN) [18] to propagate information between content nodes and hyperedges. Through this process, we obtain the node features X_N and hyperedge features X_E.

Hyperedge Association Network Construction. Traditional hypergraph modeling typically treats hyperedges merely as intermediate structures for message passing among nodes, rather than as independent semantic entities. As a result, such approaches fail to explicitly model the interactions and dependencies among different behaviors [24]. In cyberbullying, coordinated group behaviors are often closely related and mutually reinforcing within a session. To explicitly model these behaviors relationship, we propose the Interaction Behavior Graph (IBG) denoted as $G_{IB} = \{E_H, E_G\}$, which reinterprets each hyperedge $E_i \in E_H$ as a group behavior node. The edges $e_{ij} \in E_G$ model the interaction strength between different behaviors, enabling the model to explicitly reason about how group behaviors relate to each other within a session.

Given the collective nature of cyberbullying, interaction behaviors that share participants tend to exhibit strong associations [5]. Accordingly, we quantify the interaction strength between two hyperedges E_i and E_j using Jaccard similarity, which serves as the association weight W_{ij}:

$$W_{ij} = \frac{|V_i \cap V_j|}{|V_i \cup V_j|}, \tag{5}$$

where V_i and V_j denote the sets of content nodes participating in hyperedges E_i and E_j, respectively.

Collaborative Pattern Feature. We use the hyperedge representations X_E learned by the HGNN as initial node features of the IBG and apply a Graph Attention Network (GAT) to propagate information among behavior nodes to obtain updated representations. To identify groups of closely related interaction behaviors in the IBG, we apply the Louvain community detection algorithm [25], which partitions the graph into interaction behavior communities with dense connectivity and high participant overlap. We then perform mean pooling within each community to obtain community-level representations C_{comm}. These representations capture community-level collaborative patterns.

C. Multi-dimensional Information Fusion

In this module, we employ a multi-dimensional attention fusion mechanism to integrate information across nodes, hyperedges, and communities. Specifically, we first obtain the global collaborative pattern C_G through attention pooling over the community-level features C_{comm}. We then concatenate node and hyperedge features to form $X_L = \{X_N; X_E\}$. Conditioned on C_G, an attention mechanism highlights local content nodes and group behaviors that are most consistent with the global collaborative pattern. Finally, the attended local features are fused with C_G through a residual connection to produce the final session representation C_{final} as follows:

$$C_L = \text{softmax}\left(\frac{C_G X_L^\top}{\sqrt{d_k}}\right) X_L, \tag{6}$$

$$C_{\text{final}} = C_G + \text{Linear}(C_L), \tag{7}$$

where d_k is the scaling factor, $\text{softmax}(\cdot)$ normalizes the attention scores into a probability distribution over the local sequence, and $\text{Linear}(\cdot)$ is a learnable projection layer.

Session Classification. Since the imbalanced class often leads to systematic bias in the learned decision boundary and favors majority classes in the classifier's logits, we adopt the logit technique proposed by Menon et al. [26] to compensate for the skewed training data distribution. Specifically, the adjusted logits are computed as:

$$\text{Logits}_i = f(C_{\text{final}}) - \delta \cdot \log \pi, \tag{8}$$

where $f(\cdot)$ is the classification function, π represents the class frequency, and δ is a learnable scaling parameter that enables dynamic adjustment for each class.

In the task of cyberbullying detection, session-level annotations often involve subjective judgment, and manually assigned labels may contain noise. To enhance the robustness of the model, we leverage the confidence scores that refer to the consistency of annotators to construct a confidence-weighted cross-entropy loss. The loss is defined as follows:

$$\mathcal{L}(\theta) = \frac{1}{B} \sum_{i=1}^{B} \text{Confidence}_i \cdot \mathcal{L}_{\text{CE}}(\text{Logits}_i, Y_i), \tag{9}$$

where B is the total number of sessions, $\mathcal{L}_{\text{CE}}$ is the standard cross-entropy loss, Confidence_i denotes the consistency of annotators for session S_i, $Y_i \in \{0, 1\}$ represents the ground-truth label of the media session, and θ represents all parameters in the network.

V. Experiments

A. Dataset

We conduct experiments on two public datasets, Instagram [14] and Vine [15]. These datasets are widely recognized and used in media-session-based cyberbullying studies.

B. Experimental Setup

1) Evaluation Metrics: We use five standard evaluation metrics for comparisons among cyberbullying detection models [5], [6], [9]: Accuracy, F1, Precision, Recall, and AUC.

2) Baseline Methods: To comprehensively evaluate the effectiveness of our approach, we compare it against representative baselines that model cyberbullying detection from three mainstream modeling paradigms: (1) sequence-based session modeling methods: HANCD [9] and LS-CB [27], (2) graph-based session modeling methods: GNN [4], HENIN [5] and TGBully [6], (3) hypergraph-based session modeling methods: HGNN [18], HGFND [12] and HA-GCEN [23].

3) Parameter Settings: The semantic-temporal coordinated behavior hyperedge construction thresholds θ are set to 0.65 and 0.6 for the Instagram and Vine datasets, respectively. During the hypergraph construction process, we set semantic sensitivity τ to 0.1, semantic compensation η to 0.55, and temporal weight β to 0.5. We train the model for 40 epochs with a learning rate of 10^{-4} and dropout of 0.2. We split the dataset into 80% for training and 20% for testing. We report results averaged over five runs.

C. Experimental Results

As shown in Table I, the proposed BeHM consistently achieves the best performance on both datasets. Based on these results, we make the following key observations:

1) In contrast to sequence-based methods, which rely on attention mechanisms and sliding windows to model temporal dependencies, BeHM achieves a better balance between precision and recall. While sequence-based models struggle to capture complex user relationships, BeHM explicitly models relational structures among users and their interactions via hypergraphs.

2) In contrast to graph-based approaches, which model interactions using binary relations, BeHM exhibits superior overall detection performance. Graph-based methods are limited to modeling pairwise user relationships, which restricts

TABLE I Performance comparison among different models on the Instagram and Vine datasets.

Methods	Instagram					Vine				
	Acc.	**Prec.**	**Rec.**	**F1**	**AUC**	**Acc.**	**Prec.**	**Rec.**	**F1**	**AUC**
HANCD [9]	0.740	0.583	0.647	0.598	0.715	0.733	0.643	0.482	0.499	0.664
LS-CB [27]	0.846	0.722	**0.820**	0.768	0.908	0.778	0.629	0.721	0.672	0.827
GNN [4]	0.849	0.801	0.697	0.734	0.896	0.790	0.679	0.623	0.646	0.817
TGBully [5]	0.847	0.731	0.707	0.719	0.811	0.743	0.651	0.513	0.613	0.684
HENIN [6]	0.841	0.774	0.732	0.738	0.818	0.783	0.679	0.607	0.637	0.735
HGNN [18]	0.864	0.805	0.749	0.773	0.906	0.802	**0.710**	0.627	0.664	0.828
HGFND [12]	0.864	0.812	0.734	0.769	0.905	0.797	0.687	0.672	0.674	0.844
HA-GCEN [23]	0.866	0.789	0.778	0.783	**0.915**	0.798	0.666	0.712	0.688	0.838
BeHM (Ours)	**0.876**	**0.814**	0.784	**0.796**	0.907	**0.816**	0.702	**0.738**	**0.716**	**0.845**

their ability to represent coordinated group behaviors involving multiple participants, leading to inferior F1-scores.

3) In contrast to existing hypergraph-based methods, BeHM further advances high-order relational modeling by treating hyperedges as semantically meaningful group behaviors rather than auxiliary aggregation structures. While prior methods overlook interactions among hyperedges within a session, BeHM explicitly models relationships among group behaviors at the macro level to capture their collaborative patterns within a session.

As a result, BeHM achieves F1-scores of 0.796 on Instagram and 0.716 on Vine, surpassing the strongest baseline by 1.3% and 2.8%, respectively, while maintaining consistently high recall across datasets.

D. Ablation Study

To evaluate the contribution of each key component, we conduct an ablation study with five model variants. BeHM-STC, BeHM-User, and BeHM-Ment, respectively, remove semantic–temporal coordinated behavior, user coherence behavior, and mention interaction behavior hyperedges in the hypergraph construction. BeHM-IBG removes the Interaction Behavior Graph, and BeHM-MDF removes the multi-dimensional fusion. Table II reports the distribution of different hyperedge types across datasets. We observe that mention interaction behavior hyperedges account for only a very small proportion in the Vine dataset. Consequently, BeHM-Ment is excluded from the ablation study on Vine.

TABLE II Hyperedge Distribution in Instagram and Vine.

Dataset	STC	User	Ment
Instagram	137153 (85.57%)	12922 (8.06%)	10216 (6.37%)
Vine	63389 (91.71%)	5704 (8.25%)	23 (0.03%)

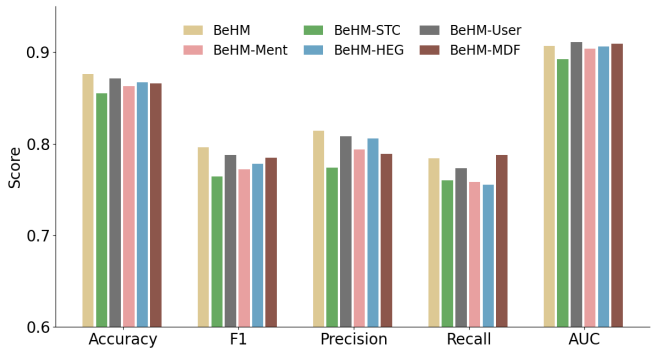
The results are shown in Fig. 3. The ablation study of BeHM provides the following insights:

1) All components contribute to the model performance. Removing any component degrades the results, confirming that each module is essential.

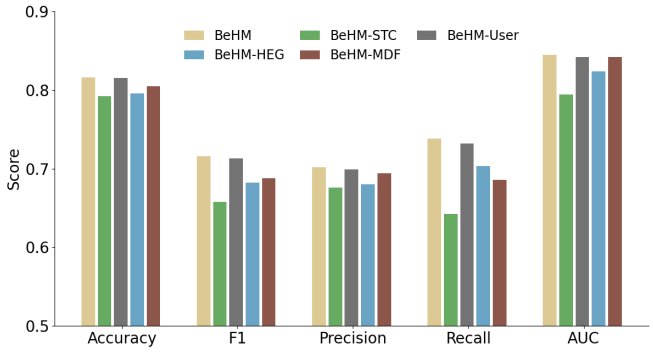
2) Among the three hyperedge types, removing semantic-temporal coordinated behavior hyperedges (-STC) causes the largest performance drop, indicating that cyberbullying primarily manifests through implicit collective attacks.

3) Removing the Interaction Behavior Graph (-IBG) decreases performance on both datasets, confirming that coordinated group behaviors are interdependent rather than isolated within sessions.

4) Removing the multi-dimensional fusion mechanism (-MDF) also reduces performance, with a more pronounced effect on Vine. This is likely due to the fragmented nature of text on Vine. The local features contain substantial noise, and without alignment to the global session representation, the model struggles to identify true cyberbullying patterns from complex behaviors.



(a) Instagram



(b) Vine

Fig. 3 Results of ablation studies for different BeHM variants.

E. Case Study

In this section, a case study is used to verify the interpretability and effectiveness of the BeHM model for identifying multi-level cyberbullying patterns. Fig. 4 presents a representative cyberbullying session and the top-5 comments highlighted by BeHM based on fused attention scores. BeHM assigns high weights to a sequence of temporally concentrated and highly aggressive replies, mostly authored by the same user (User3) and directly targeting another participant (User8). This indicates that BeHM not only captures individual offensive language but also recognizes temporally concentrated and

user-consistent behaviors as a coordinated attack pattern. In addition, the highlighted comments revolve around the first comment, suggesting that BeHM effectively identifies the core content driving the escalation of cyberbullying in the session.

Original Posted text	**Label**
Ooohhh kill em.... My girl Laura making me feel like a star.	(#Cyberbullying = True)

User	Time	Top-5 Comments Ranked by BeHM
User3	10-18 19:00	*STOP WITH THE OOOH KILL EM*
User3	10-18 19:13	*@user8 how about you shut the fuck up and mind your business bitch? Who are you? Ciph's bodyguard. Loser*
User3	10-18 19:14	*@user8 and its actually an annoying ass phrase, not a word you dumb fuck*
User8	10-18 19:20	*@user3 Buuhuu.. Please go cry somewhere else, cuntface!*
User3	10-18 19:22	*@user8 you're the one looking for attention. Now go eat a cookie and make a twerk video you useless twat*

Fig. 4 Top-5 comments highlighted by BeHM.

VI. Conclusion

In this paper, we investigate the collective behavioral patterns underlying cyberbullying and the limitations of existing sequential and pairwise graph methods in modeling high-order interactions. To address these limitations, we propose BeHM, a behavior-driven hypergraph framework. BeHM captures coordinated attack behaviors at the micro level and their relationship at the macro level through hyperedge construction, interaction behavior graph modeling, and multi-dimensional information fusion. Experimental results on two public datasets show that BeHM consistently outperforms existing methods. These results demonstrate the effectiveness of high-order relational modeling and layered multi-dimensional fusion for session-level cyberbullying detection. Future work can extend this framework to finer-grained behavior analysis, cross-session relation modeling, and multimodal integration.

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